Pranay Reddy Ragi

me23btech11049@iith.ac.in | ragipranay0803@gmail.com | +91 7702360623 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.97
Sri Chaitanya Junior College	Class XII (TSBIE)	2023	98.60%
Johnson Grammar School	Class X (ICSE)	2021	97.40%

WORK EXPERIENCE

Dunkit | *Applied AI Engineer*

Japanese technology startup focused on AI-driven unmanned stores and SaaS solutions

AI CCTV Analytics

- Developed a multi-stage CCTV analytics pipeline combining YOLO-based detection, persistent person tracking, rule-based suspicious-behavior analysis and automated pre/post-event clip generation for evidence.
- Generated 500+ suspicious candidate clips and 1,400+ person clips from ~30-40+ CCTV videos, and trained a PyTorch normal-vs-suspicious classifier to validate suspicious-event candidates.

QR-Based Device Activation System

- Integrated a USB HID QR scanner with Raspberry Pi using Python evdev to process device IDs and publish device-specific MQTT activation commands to ESP32.
- Validated the end-to-end QR-to-device-control workflow across Raspberry Pi, Mosquitto MQTT and ESP32, successfully triggering a physical LED response upon QR-based activation.

SKILLS

Languages: C++, Java, Python, JavaScript, SQL

Web Technologies: HTML, CSS, React, Node.js, Express.js, MongoDB

CS Fundamentals: Data Structures & Algorithms, OOP, DBMS

Tools: Git, GitHub

PROJECTS

Student-Teacher Appointment Booking System

- **Built a full-stack appointment platform using React, Node.js, Express and MongoDB**, enabling students to browse teachers, view available slots and book appointments.
- **Implemented JWT-based authentication with role-specific dashboards** for students, teachers and admins, supporting slot management, appointment requests and approval workflows.
- **Developed appointment tracking and email notifications**, allowing users to view upcoming and previous bookings while receiving updates on appointment status.

Smart Ad & Tracker Blocker

- Developed a **Manifest V3 Chrome extension** with network-level request interception to block advertising and analytics trackers using configurable domain-based rules.
- Implemented **persistent user preferences and blocking controls** with Chrome Storage, enabling customizable filtering and management of blocked requests.
- Added **real-time blocking statistics** to track intercepted advertising and analytics requests and provide visibility into blocking activity.

Airline Flight Operations & Delay Analytics

- Engineered a Python pipeline to process **7M+ U.S. flight records**, performing chunked data ingestion, cleaning, validation and feature engineering for large-scale analysis.
- Built **DuckDB SQL-based analytics** to generate airline, airport, route and time-based KPIs, uncovering patterns in delays, cancellations and diversions across flight operations.
- Identified delay rates rising from **8.9% at 5 AM to 29.8% at 8 PM**, with late-aircraft and carrier-related issues contributing **~75% of total delay minutes**.

Job Eligibility & Skill Matching System

- Built a web-based eligibility engine that parses JDs to extract **CGPA, branch, graduation year, degree, skills, and experience requirements**, enabling automated candidate eligibility and skill-gap analysis.
- Developed an **explainable matching engine and Flask dashboard** with PDF processing and SQLite-backed analysis history to provide clear reasons for candidate fit or rejection.

POSITIONS OF RESPONSIBILITY

Placement Coordinator, Office of Career Services (**OCS**), IIT Hyderabad

Student Representative, National Service Scheme (**NSS**), IIT Hyderabad

Operations Head, Ek Bharat Shreshtha Bharat (**EBSB**), IIT Hyderabad

EXTRACURRICULAR

Ranked 1st among IIT Hyderabad participants in Naukri's Engineers' Ring of Honour (Nov 2025).

Enjoy playing **Matiks**, particularly its mathematical and puzzle-solving challenges.

Proposed and co-organized Sign Language, CPR, and First Aid workshops through NSS at IIT Hyderabad, collaborating with the team to execute campus-wide sessions.